

Discrete Logarithm ¶

The discrete logarithm is an integer x satisfying the equation

$$a^x \equiv b \pmod{m}$$

for given integers a , b and m .

The discrete logarithm does not always exist, for instance there is no solution to $2^x \equiv 3 \pmod{7}$. There is no simple condition to determine if the discrete logarithm exists.

In this article, we describe the **Baby-step giant-step** algorithm, an algorithm to compute the discrete logarithm proposed by Shanks in 1971, which has the time complexity $O(\sqrt{m})$. This is a **meet-in-the-middle** algorithm because it uses the technique of separating tasks in half.